
When Does Neural-Symbolic Integration *Understand*? Data Efficiency, Reasoning Shortcuts, and Compositional Generalization on MNIST-Addition

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Abstract

Neuro-symbolic (NeSy) systems promise to combine neural perception with symbolic reasoning, yielding data efficiency, interpretable symbols, and compositional reuse. But these promises rest on a hidden assumption: that the system actually *grounds* its latent symbols correctly. We ask, on the canonical MNIST-ADDITION benchmark, when wiring a fixed symbolic “+” into a shared neural digit classifier—trained only on the *sum* of two images, never on the digits themselves—produces genuine perception-plus-reasoning, and when it only appears to. Across 210 training runs (5 seeds per configuration), we report four findings measured side by side with confidence intervals and significance tests. First, integration is a large data-efficiency win: at 1,000 training pairs the NeSy model reaches 85.4% sum accuracy versus 22.0% for a pure-neural baseline (Welch $p=2.9\times 10^{-13}$), matching a fully digit-supervised oracle. Second, high task accuracy can hide a broken model: changing only the symbolic operator to $(a+b) \bmod 10$ makes correct grounding *never* emerge—raw grounding $\approx 0.1\%$ among seeds that learn the task—because the network reads digits via a systematic relabeling that leaves the modular sum invariant, a reasoning shortcut visible only when grounding is measured permutation-aligned. Third, just 10 labeled images (one per class) eliminate the shortcut and stabilize training (grounding $\approx 0.1\% \rightarrow 97.7\%$), whereas unsupervised regularizers do not. Fourth, correct grounding unlocks zero-shot compositional generalization to 2- and 3-digit addition (95.7%/93.7%) that the neural baseline structurally cannot achieve (0%). The lesson for practitioners: task accuracy is not enough; grounding must be measured, the operator’s symmetry determines whether symbols are identifiable, and a pinch of symbol supervision is a cheap, decisive fix.

1 Introduction

Neuro-symbolic (NeSy) AI sets out to build a single system that *perceives* with neural networks and *reasons* with symbolic structure [Garcez and Lamb, 2023, Marra et al., 2024]. The appeal is concrete: by factoring a task into “recognize the parts” and “combine them with known rules,” a NeSy model should learn from far less data, expose interpretable intermediate symbols, and reuse those symbols to solve problems it was never trained on. The canonical demonstration is MNIST-ADDITION [Manhaeve et al., 2018]: a model sees two digit images and is supervised only on their *sum*, never on the digits, yet is expected to learn to read digits as a by-product of learning to add.

The promise hides an assumption. Every downstream NeSy benefit—interpretability, knowledge reuse, compositional generalization—depends on the latent symbols being *correct*. If the network learns a mapping from images to symbols that is systematically wrong but still produces the right final answer, the system has not understood anything; it has found a **reasoning shortcut** [Li et al.,

2024, Daniele et al., 2022]. A practitioner who watches only task accuracy will never notice. This is not a hypothetical: with only distant supervision, the perception-to-symbol map is under-constrained, and training can collapse onto a spurious but self-consistent solution.

What is missing. The two headline NeSy claims—data efficiency and correct grounding—are widely asserted but rarely measured together with statistical rigor on identical splits. Data efficiency is seldom reported head-to-head against a neural baseline with confidence intervals, and *grounding accuracy is almost never reported at all*, so reasoning shortcuts stay invisible. Prior work raises the shortcut problem qualitatively, but a controlled, multi-seed *measurement of when* shortcuts appear, *what they look like*, and *which cheap fixes work* has been missing.

Our approach. We run a single controlled study on MNIST-ADDITION that measures four quantities side by side, on identical splits, across 5 seeds, with Welch t -tests, Holm–Bonferroni correction, and Cohen’s d effect sizes. We compare three models on a *shared* CNN encoder so the only variable is the learning signal: a NEURAL-ONLY baseline that predicts the sum directly, a NESY model that outputs per-image digit distributions and computes the sum by exact differentiable marginalization (the DeepProbLog/Scallop idea in compact form), and an ORACLE trained on true digit labels as an upper bound. The study is held together by one clean manipulation: *flipping a single symbolic operator*, standard addition $a+b$ versus modular addition $(a+b) \bmod 10$, which turns reasoning shortcuts on and off while everything else—architecture, data, optimizer—stays fixed.

What we find. (i) Integration is a large, statistically significant data-efficiency win: at 1,000 pairs the NESY model reaches 85.4% sum accuracy versus the neural baseline’s 22.0% (Cohen’s $d=56$, $p=2.9 \times 10^{-13}$), recovering digits at 92–99% with no digit labels. (ii) Yet under the modular operator, correct grounding *never* emerges: among seeds that learn the task, raw grounding is $\approx 0.1\%$, while a permutation-aligned metric reveals a clean systematic relabeling—sometimes the exact cyclic shift $d \mapsto (d+5) \bmod 10$. The oracle grounds at 99% on the *same* task, so the failure is caused by distant supervision under a symmetric operator, not by the task being hard. (iii) Just 10 labeled images break the symmetry and restore grounding to 97.7% in every seed; unsupervised regularizers do not. (iv) With correct grounding, the NESY model solves 2- and 3-digit addition it never saw at 95.7%/93.7% by composing its digit classifier with a symbolic place-value sum, while the neural baseline scores exactly 0%.

Contributions.

- We quantify NeSy data efficiency head-to-head against a neural baseline and a digit-supervised oracle on identical splits, with confidence intervals and effect sizes at six training sizes.
- We introduce a permutation-aware (Hungarian-aligned) grounding diagnostic and use a single-operator flip ($a+b$ vs. $(a+b) \bmod 10$) to show that whether high task accuracy implies correct symbols is a property of the *task’s symmetry*, not of NeSy in general.
- We compare cheap mitigations and show that a handful of labeled symbols (10 images) is the decisive, reliable fix for a detected shortcut, while unsupervised priors are not.
- We demonstrate that correct grounding unlocks zero-shot compositional generalization to multi-digit addition—a capability the neural baseline structurally lacks.

The rest of the paper reviews related work (section 2), details our models, operators, and metrics (section 3), presents results E1–E5 (section 4), discusses interpretation and limitations (section 5), and concludes (section 6).

2 Related Work

Probabilistic logic with neural predicates. A major NeSy family compiles logic into a differentiable circuit and lets a neural network supply the probabilities of atoms. DeepProbLog [Manhaeve et al., 2018] adds a neural predicate to ProbLog, compiles inference to arithmetic circuits, and introduces MNIST-ADDITION as its flagship benchmark; NeurASP [Yang et al., 2020] couples network softmax outputs to an answer-set program; and Scallop [Li et al., 2023] implements differentiable Datalog with provenance semirings and a tunable top- k approximation for scale. Our NESY model is a compact re-implementation of the shared core of these systems: per-image digit distributions combined by an *exact* differentiable marginalization over digit pairs. We do not benchmark the toolkits themselves; we use their common mechanism as a clean experimental substrate and add the grounding and shortcut measurements they usually omit.

Differentiable logic as loss or representation. A second family treats knowledge as a soft constraint on network outputs rather than as architecture. Logic Tensor Networks [Badreddine et al., 2022] define a many-valued differentiable first-order logic whose formulas become satisfiability objectives, and the semantic loss [Xu et al., 2018] measures the distance of network outputs from satisfying a logical constraint. These methods inject knowledge without changing the architecture; we instead bake the symbolic operator directly into the forward computation, which is precisely what makes the latent digit symbols inspectable and the grounding question well posed.

The symbol grounding problem and reasoning shortcuts. The closest work to ours studies how distant supervision under-constrains the perception-to-symbol map. Softened Symbol Grounding [Li et al., 2024] and Deep Symbolic Learning [Daniele et al., 2022] observe that a NeSy system can reach high task accuracy while learning a wrong, self-consistent symbol map, and propose training schemes to mitigate it. These works raise the shortcut problem largely qualitatively. We complement them with a controlled, multi-seed *measurement*: we tie shortcut emergence to a concrete, analyzable property of the symbolic operator (its symmetry), provide a permutation-aware detector, quantify the resulting collapse and its training instability, and rank cheap mitigations against one another.

Compositional and out-of-distribution generalization. Modular neuro-symbolic systems such as NS-VQA [Yi et al., 2018] and the Neuro-Symbolic Concept Learner [Mao et al., 2019] achieve strong compositional generalization by pairing a neural perception front-end with a symbolic executor, and multi-digit/multi-addend addition is a standard stress test for inference scaling [Manhaeve et al., 2018, Li et al., 2023]. We use multi-digit addition differently: as a *diagnostic of grounding*. Because our single-digit classifier can be composed with a symbolic place-value sum, a model with correct grounding should generalize zero-shot while one with a shortcut should not, making OOD accuracy a direct, externally checkable test of whether the latent symbols are right.

Surveys and positioning. Broad treatments of the field [Garcez and Lamb, 2023, Marra et al., 2024] catalog the design space and repeatedly flag two gaps: data efficiency is asserted more than measured, and grounding is under-reported. Our study is deliberately narrow—one benchmark, one perceptual domain—but addresses exactly these gaps with statistics, and ties data efficiency, grounding, mitigation, and compositional generalization together through a single operator manipulation.

3 Methodology

3.1 Problem setup

MNIST-ADDITION presents pairs of digit images (x_1, x_2) with only the integer sum $s = y_1 + y_2$ supervised; the true digit labels (y_1, y_2) are hidden during training. The learning challenge is that the digit-recognition map must be discovered purely as a by-product of fitting the sum—this is what makes the benchmark a clean test of perception-via-symbolic-constraint, and what makes correct *grounding* a non-trivial outcome.

3.2 Models

All three models share an identical LeNet-style CNN encoder; the *only* difference is the output head and the learning signal, which isolates the symbolic structure as the variable.

Neural-only baseline. The shared encoder is applied to each image; the two feature vectors are concatenated and passed to an MLP that produces a 19-way softmax over possible sums $\{0, \dots, 18\}$ (a 10-way head for the modular task). It has no knowledge that the output decomposes as $d_1 + d_2$.

Neuro-symbolic (NESY). The shared encoder maps each image to a 10-way digit log-probability vector. Treating the two digits as independent given the images, the distribution over the sum is computed *exactly and differentially* as the convolution of the two digit pmfs,

$$P(\text{sum} = s) = \sum_{a+b=s} P(d_1=a) P(d_2=b), \quad (1)$$

which is the DeepProbLog/Scallop marginalization in compact form. Training minimizes the negative log-likelihood of the *true sum only* (distant supervision). The symbolic operator is fixed and never learned. For the modular task, the constraint in equation 1 becomes $a+b \equiv s \pmod{10}$.

Oracle (upper bound). The same 10-way CNN is trained with ordinary cross-entropy on the true digit labels and adds via argmax. It bounds the achievable grounding and sum accuracy and, crucially, serves as a control: because it sees digit labels, it should ground correctly *regardless* of the symbolic operator.

3.3 Symbolic operators: a controlled shortcut switch

We study two operators that differ only in their algebra. *Standard addition* ($a+b$, output 0–18) has boundary constraints—a sum of 0 forces 0+0 and a sum of 18 forces 9+9—that uniquely pin the symbols, making the grounding identifiable. *Modular addition* $(a+b) \bmod 10$ (output 0–9) deliberately removes those boundary constraints and introduces a rotational symmetry: any cyclic relabeling $d \mapsto (d+k) \bmod 10$ applied to both images leaves the modular sum invariant. The operator therefore *admits* an exact affine shortcut and is expected to *induce* one under distant supervision. The oracle is unaffected because it never relies on the sum to identify digits. This single-operator flip is the backbone of our study: it switches shortcuts on and off while holding architecture, data, and optimization fixed.

3.4 Data

We use the locally pre-built MNIST-ADDITION dataset (30,000 train pairs, 5,000 test pairs) indexing into MNIST. True digit labels are used *only* for evaluation (grounding) and for the oracle and mitigation conditions. We verified data integrity (stored sums equal the sums of the true digits; test indices map exactly to the MNIST test images). For the compositional experiment we deterministically synthesize 2- and 3-digit number-addition test sets (5,000 examples each) from MNIST test images.

3.5 Metrics

- **Test sum accuracy** — the primary task metric.
- **Digit-grounding accuracy** — *raw* (argmax digit vs. true label) and *Hungarian-aligned* (accuracy under the best 10×10 label permutation, found by solving the assignment problem). When aligned grounding greatly exceeds raw grounding, the network has learned a *systematic* relabeling—a reasoning shortcut. We flag a shortcut when aligned—raw > 0.10.
- **Compositional OOD accuracy** — multi-digit addition solved by a single-digit-trained model via symbolic place-value composition.
- **Training wall-clock** — secondary.

3.6 Protocol and reproducibility

We run 5 seeds $\{0, \dots, 4\}$ per configuration and report mean $\pm 95\%$ t -based confidence intervals. The primary test is Welch’s two-sample t -test (unequal variance) for NESY vs. NEURAL-ONLY at each training size, with Holm–Bonferroni correction across the six sizes and Cohen’s d effect sizes; for the compositional analysis we report Pearson correlation. All conditions use the same hyperparameters (12 epochs, batch size 128, Adam at 10^{-3}), set in `src/config.py`; seeds are fixed for Python/NumPy/PyTorch. The full grid ($E1+E5 = 165$ runs, $E3 = 30$, $E4 = 15$) runs in roughly five minutes across three GPUs on an NVIDIA RTX A6000 (PyTorch 2.12, CUDA). Raw per-run results, the aggregate statistics, and figures are saved under `results/` and `figures/`.

4 Results

We organize results around the four hypotheses: data efficiency (E1), grounding and its silent failure under a flipped operator (E2/E5), mitigations (E3/E5), and compositional generalization (E4).

4.1 E1 — Data efficiency: integration is a large, significant win at every size

Table 1 and figure 1(a) compare the three models across six training sizes on standard addition. The NESY model beats the NEURAL-ONLY baseline at *every* size (all Holm-adjusted $p < 0.01$), tracks the oracle closely, and the gap is largest in the low-data regime. The single most striking point is at 1,000

# train pairs	NEURAL-ONLY	NESY (sum-only)	ORACLE	NESY – NEURAL-ONLY	Cohen’s d	Welch p (raw)	Holm-adj. p
100	0.086	0.130	0.429	+0.044	2.28	8.8×10^{-3}	8.8×10^{-3}
500	0.130	0.649	0.794	+0.519	5.43	9.3×10^{-4}	1.9×10^{-3}
1,000	0.220	0.854	0.868	+0.634	56.3	2.9×10^{-13}	1.7×10^{-12}
5,000	0.883	0.955	0.958	+0.072	10.25	1.6×10^{-6}	7.9×10^{-6}
15,000	0.952	0.973	0.975	+0.021	4.82	3.8×10^{-4}	1.5×10^{-3}
30,000	0.970	0.980	0.981	+0.010	4.13	4.8×10^{-4}	1.5×10^{-3}

Table 1: **E1: data efficiency on standard addition.** Test sum accuracy (mean over 5 seeds); NESY is the sum-only neuro-symbolic model, ORACLE is digit-supervised. NESY beats the NEURAL-ONLY baseline at every training size, with the largest gap at 1,000 pairs ($d=56.3$). Every Holm-adjusted $p < 0.01$, so all comparisons remain significant after family-wise correction. Best of {NEURAL-ONLY, NESY} in **bold**.

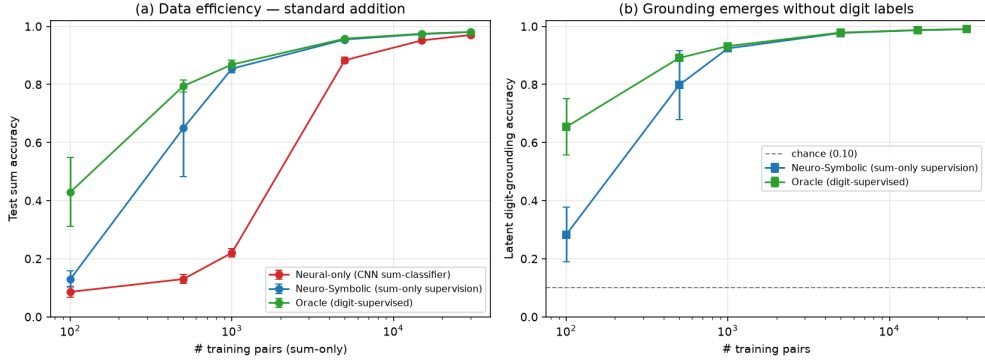


Figure 1: (Left) Data efficiency on standard addition. NESY (blue) dramatically beats the NEURAL-ONLY baseline (red), most sharply in the low-data regime, and tracks the digit-supervised oracle (green); the baseline needs $\sim 5\times$ more data to catch up. (Right) Latent digit-grounding accuracy rises to 99% for NESY with *no* digit labels, matching the oracle and far above the 0.10 chance line. Error bars are 95% confidence intervals over 5 seeds.

pairs, where NESY reaches 85.4% versus the baseline’s 22.0%—a Cohen’s d of 56 ($p=2.9 \times 10^{-13}$). The neural baseline needs roughly 5,000 pairs to reach what NESY achieves with about 1,000. As data grows the gap shrinks but never closes (e.g. +1.0 point at 30,000, still $p < 0.01$), exactly the pattern the data-efficiency claim predicts.

4.2 E2 — Grounding emerges without any digit labels (standard addition)

figure 1(b) shows that NESY’s latent digit-grounding rises from 28.3% (100 pairs) to 99.0% (30k pairs), essentially matching the digit-supervised oracle—even though NESY never saw a digit label. The Hungarian-aligned grounding *equals* the raw grounding at every size (0/5 shortcut flags throughout), so standard MNIST-ADDITION is **shortcut-robust**: its boundary sums uniquely pin the symbols. The visible sum-vs-grounding gap (e.g. 85.4% sum vs. 92.4% grounding at 1k) is just the “both digits must be right” effect (sum $\approx$ grounding²), not a shortcut.

4.3 E5 — Flip the operator and grounding collapses

Changing *only* the symbolic operator to $(a+b) \bmod 10$ produces a sharp dissociation between task accuracy and grounding (Table 2, figures 2 and 3). Standard addition grounds at 99.0%; modular addition grounds at $\approx 2\%$ raw. Among the four seeds that actually learn the modular task (sum > 0.3), raw grounding is just 0.09% (max 0.16%): the network reads the digits systematically wrong via a global relabeling that leaves $(a+b) \bmod 10$ invariant. The permutation-aligned metric makes the failure legible—aligned grounding is 0.53–0.60 on shortcut seeds and reaches 0.99 on the cleanest realization, where the model lands on the *exact* predicted cyclic shift $d \mapsto (d+5) \bmod 10$ (figure 3). The fifth seed collapses to predicting a single class. Crucially, the digit-supervised *oracle grounds at 99.0% on the identical modular task*, so the collapse is caused by distant supervision under a symmetric operator, not by the task. The standard-vs-modular grounding contrast is overwhelming

Task (NESY, $N=30k$)	Sum acc (mean $\pm$ CI)	Grounding (raw)	Hungarian-aligned	Outcome over 5 seeds
standard $a+b$	0.980 ± 0.002	0.990	0.990	0/5 shortcut (grounds correctly)
modular $(a+b) \bmod 10$	0.657 ± 0.380	0.020	0.471	4/5 shortcut + 1/5 collapse
modular — ORACLE (digit-sup.)	0.979	0.990	0.990	0/5 (control: same task, grounds fine)

Table 2: **E5: flipping the operator switches shortcuts on.** Changing only $a+b \rightarrow (a+b) \bmod 10$ leaves task accuracy non-trivial but collapses raw grounding from 0.990 to 0.020; the large aligned—raw gap reveals a systematic relabeling. The digit-supervised oracle grounds at 0.990 on the *same* modular task (control), so the failure is caused by distant supervision under a symmetric operator. Standard vs. modular grounding: Welch $p=9.6 \times 10^{-7}$, $d=31.5$.

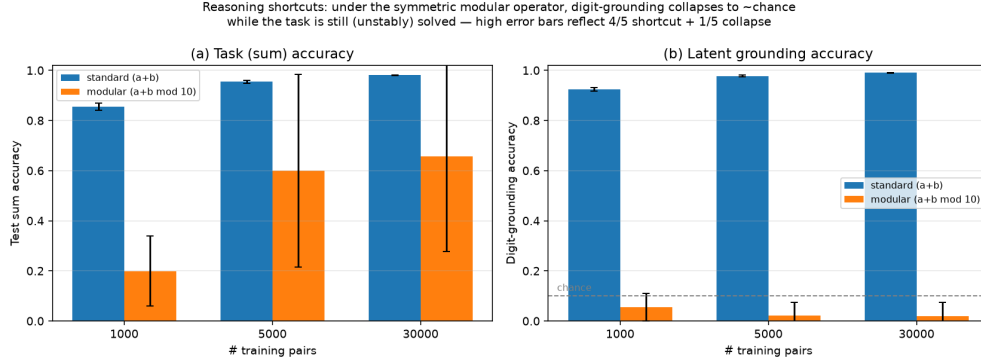


Figure 2: Flipping one operator switches reasoning shortcuts on. (*Left*) Task (sum) accuracy: standard addition (blue) is high and stable, while modular addition (orange) is solved *unstably*—the large error bars reflect 4/5 seeds finding a shortcut and 1/5 collapsing. (*Right*) Latent grounding: standard addition reaches 99%, but modular addition sits near the 0.10 chance line—high task accuracy with near-zero grounding. Bars are means over 5 seeds; error bars are 95% confidence intervals.

(0.990 vs. 0.020, Welch $p=9.6 \times 10^{-7}$, $d=31.5$). This shortcut is visible *only* because we measured grounding; task accuracy alone is blind to it.

4.4 E3/E5 — Mitigations: needed and effective exactly when a shortcut exists

Table 3 and figure 4 report grounding accuracy under each intervention. We evaluate standard addition at $N=1k$ (where its grounding gap is largest) and modular addition at $N=5k$ (where the shortcut is firmly established); each column is a within-task before/after comparison, so the differing N does not affect the conclusion. On standard addition, where there is no shortcut to fix, mitigations do essentially nothing. On modular addition, *just 10 labeled images* (one per class) break the symmetry and restore grounding from $\approx 0.1\%$ to 97.7% in *every* seed (per-seed 0.973–0.981; Welch $p=1.6 \times 10^{-3}$, $d=4.8$), while also stabilizing training (sum accuracy $\rightarrow 0.954$, no collapses). Adding more labels (50, 200) does not help further. Unsupervised regularizers are unreliable: a marginal-uniformity prior recovers only partial grounding (0.390, high variance) and an entropy penalty does not help at all (0.128). The cure matches the disease—supply the missing identifying constraint—and it is extremely cheap.

4.5 E4 — Compositional generalization: the payoff of correct grounding

Finally, Table 4 and figure 5 show the capability that correct grounding unlocks. Trained on *single-digit* sums only, the NESY model solves 2- and 3-digit number addition it never saw—at 95.7% and 93.7%—by composing its digit classifier with a symbolic place-value sum, essentially matching the oracle (96.0%/94.0%). The NEURAL-ONLY baseline scores exactly 0%: it cannot even emit sums in this range, a structural rather than a tuning failure. Across the grounded models, single-digit grounding and OOD accuracy correlate almost perfectly (Pearson $r=0.993$, $p=1 \times 10^{-8}$), but because that correlation spans models that all ground well, we treat the *categorical* contrast (0% for the ungrounded baseline vs. $\sim 96\%$ for grounded models) as the stronger evidence. This is the most

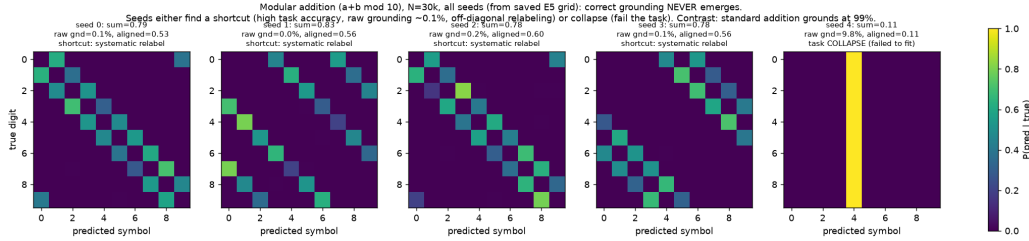


Figure 3: Per-seed grounding confusion matrices (rows: true digit, columns: predicted symbol) for modular addition at $N=30k$, read from the saved E5 grid. Seeds 0–3 place nearly all mass *off* the diagonal: a clean, systematic relabeling (a reasoning shortcut) rather than random noise—seed 1 is close to the exact $d \mapsto (d+5)$ cyclic shift. Seed 4 collapses to a single predicted class (failed to fit). Correct grounding never appears, whereas standard addition grounds on the diagonal at 99%.

Intervention	Standard add @ $N=1k$ (no shortcut)	Modular add @ $N=5k$ (shortcut present)
baseline (none)	0.924	0.116 (4/5 ≈ 0 ; 1 seed 0.57)
+ aux digit labels, 1/class (10 total)	0.925	0.977
+ aux 5/class (50 total)	0.926	0.977
+ aux 20/class (200 total)	0.931	0.979
+ marginal-uniformity prior	0.924	0.390
+ entropy regularization	0.925	0.128

Table 3: **E3: mitigations restore grounding only when a shortcut exists.** Digit-grounding accuracy under each intervention (standard addition evaluated at $N=1k$, modular at $N=5k$; each column is a within-task before/after). On standard addition there is nothing to fix. On modular addition, just 10 labeled images break the symmetry and restore grounding to 0.977 in every seed (Welch $p=1.6 \times 10^{-3}$, $d=4.8$), whereas unsupervised regularizers help only partially (uniform prior) or not at all (entropy). Best modular result in **bold**.

direct demonstration that integration buys a genuinely new capability, not just a higher benchmark number—and that the capability rests on correct grounding.

5 Discussion

Integration is a real capability and data-efficiency win. Wiring the symbolic operator into the model factorizes the task into digit recognition, so NeSy learns from far fewer examples and matches a fully supervised oracle (section 4.1). Effect sizes are large at every size and overwhelming at 1,000 pairs ($d=56$). We provide what the literature usually only asserts: a head-to-head learning-curve comparison with confidence intervals on identical splits.

Whether high accuracy implies correct symbols is a property of the operator, not of NeSy. This is our central and most useful finding. Standard addition is shortcut-robust because its boundary sums identify the digits; modular addition is shortcut-prone because its rotational symmetry does not. The same network swings from 99.0% grounding to $\approx 0.1\%$ (among task-learning seeds) when we flip a single operator, while the oracle—which sees digit labels—grounds at 99% on *both* operators, the clean control (section 4.3). The Hungarian-aligned metric is what makes the failure legible: aligned grounding far above the raw value exposes a *systematic* relabeling rather than noise, and in the cleanest case the model lands on the exact algebraically predicted cyclic shift. The methodological implication is direct: report grounding, and report it permutation-aligned; sum accuracy alone is blind to this failure.

The cure matches the disease. When the problem is non-identifiability, supplying a tiny amount of identifying information—10 labeled images, one per class—restores grounding to 97.7% in every seed and also removes the training instability (section 4.4). Unsupervised regularizers, which add no identifying information, mostly cannot. This reframes “mitigation” as “supply the missing constraint” and shows it is cheap when needed and unnecessary when not.

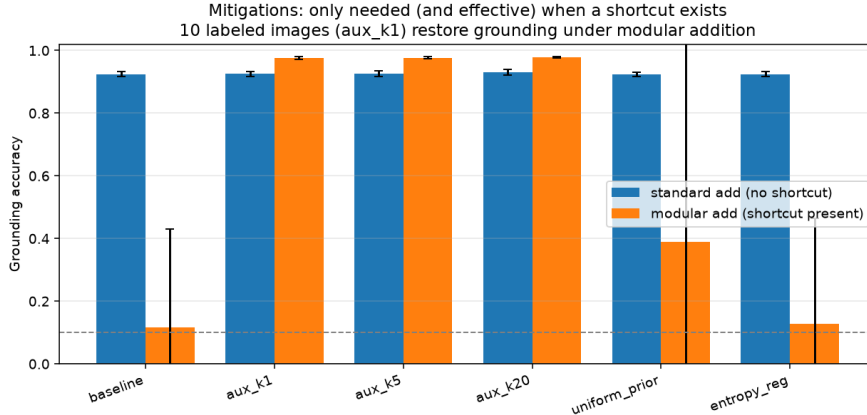


Figure 4: Mitigations are needed (and effective) only when a shortcut exists. On standard addition (blue, no shortcut) all interventions leave grounding essentially unchanged. On modular addition (orange, shortcut present) just 10 labeled images (aux_k1) restore grounding to 97.7%; more labels add nothing, the uniform prior helps only partially (high variance), and the entropy penalty does not help. Dashed line is the 0.10 chance level.

Model	2-digit (OOD)	3-digit (OOD)	single-digit grounding
NEURAL-ONLY	0.000	0.000	0.000 (n/a)
NESY	0.957	0.937	0.990
ORACLE	0.960	0.940	0.990

Table 4: **E4: compositional generalization.** Models trained only on single-digit sums, evaluated zero-shot on multi-digit number addition via symbolic place-value composition. NESY nearly matches the oracle and generalizes to a structure it never saw, *because* its grounding is correct; the NEURAL-ONLY baseline cannot represent outputs in this range (structural 0%). Best of the two learnable contrasts (NEURAL-ONLY, NESY) in **bold**.

Correct grounding is what compositional generalization stands on. The single-digit-trained NESY model solves multi-digit addition at $\sim 94\text{--}96\%$ because, and only because, its digit classifier is correct; the neural baseline’s structural 0% is the foil (section 4.5). We are careful here: the near-perfect correlation between grounding and OOD accuracy spans models that all ground well and has little spread, so we lean on the categorical contrast (0% vs. $\sim 96\%$) rather than over-reading the correlation.

Surprises. Two results were sharper than we expected. First, a single operator flip makes correct grounding vanish so reliably—raw grounding $\approx 0.1\%$ in every seed that learns the task—and the model occasionally realizes the *exact* $d \mapsto (d+5)$ permutation, together with training instability (4/5 shortcut, 1/5 outright collapse). Second, the neural baseline shows high variance under modular addition at 30k (four seeds at 0.96–0.97, one stuck at 0.10), hinting that the modular task is genuinely harder to represent without the additive factorization.

Error analysis. NESY’s residual errors on standard addition fall on genuinely ambiguous digit images, consistent with its $\sim 99\%$ grounding—they are *perceptual*. Its “errors” under modular addition are not perceptual at all: they stem from a systematic global relabeling of the symbols (figure 3), which is precisely why high task accuracy survives them.

5.1 Limitations

We are explicit about scope. (i) *Single benchmark, single perceptual domain:* all results are on MNIST digits; the mechanism (task symmetry $\Rightarrow$ (non)identifiability of symbols) should transfer, but we did not test other modalities such as HWF or CLEVR. (ii) *One symbolic relaxation:* our NESY model is a faithful compact re-implementation of the DeepProbLog/Scallop exact marginal-

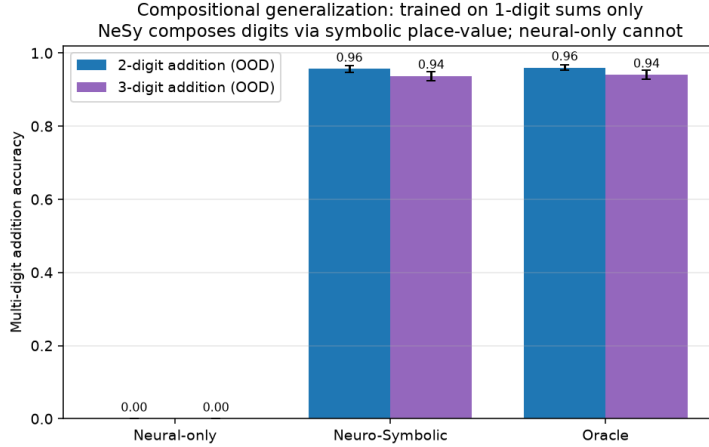


Figure 5: Zero-shot compositional generalization. Trained only on single-digit sums, NESY and the oracle solve 2- and 3-digit number addition at $\sim 94\text{--}96\%$ by composing the digit classifier with a symbolic place-value sum. The NEURAL-ONLY baseline scores 0% —it structurally cannot represent outputs in this range. Error bars are 95% confidence intervals over 5 seeds.

ization, not a benchmark of those toolkits themselves. (iii) *Low-spread correlation*: the H4 correlation is suggestive only; we rely on the categorical contrast. (iv) *GPU non-determinism and modular instability*: atomic CUDA operations make runs not bit-identical even with fixed seeds, so the specific permutation a modular run lands on—and whether a seed shortcuts or collapses—varies across re-runs. The robust, re-run-invariant phenomenon is that under the modular operator correct grounding never emerges (raw grounding $\approx 0.1\%$ whenever the task is learned; the oracle always grounds at 99%). Because modular NESY is unstable, aggregate statistics over seeds, not single runs, are the unit of inference, and figure 3 reads the saved grid rather than fresh retrains. Standard-addition results reproduce across re-runs to within ± 0.01 . (v) Modular addition is a *deliberately constructed* shortcut-inducer chosen for its exact, analyzable symmetry, not a naturally occurring task.

Broader implications. For anyone deploying a NeSy system, the practical message is that task accuracy is a necessary but not sufficient signal of understanding. Latent symbols can be confidently and systematically wrong while every reported number looks healthy. Auditing grounding—and analyzing the symbolic operator for symmetries that fail to identify the symbols—should be standard practice, and where shortcuts are possible, a handful of labeled symbols is a cheap and reliable safeguard.

6 Conclusion

We asked when integrating a fixed symbolic operation with a shared neural classifier produces a system that genuinely perceives and reasons, and answered it with a single controlled study on MNIST-ADDITION. Integration (i) is a large, statistically significant capability and data-efficiency win that matches a fully-supervised oracle and beats a neural baseline, most sharply with little data; (ii) recovers the *correct* latent symbols *if and only if* the task’s structure identifies them—when it does not, as under modular addition, the system reaches high task accuracy through systematic reasoning shortcuts that only a permutation-aware grounding metric can detect; (iii) is cheaply repaired with about 10 labeled symbols exactly when a shortcut exists; and (iv) when grounding is correct, unlocks zero-shot compositional generalization to multi-digit addition ($\sim 95\%$) that the neural baseline structurally cannot achieve (0%).

Key takeaway. In neuro-symbolic systems, task accuracy is not enough: measure latent grounding permutation-aligned, analyze the symbolic operator for symmetries that fail to identify the symbols, and if shortcuts are possible add a handful of labeled symbols—the cheapest, most reliable fix, and one that unsupervised regularization does not match.

Future work. We plan to (a) replicate the symmetry $\rightarrow$ (non)identifiability mechanism on HWF and a relational task such as CLUTRR; (b) probe richer shortcut-inducing operators (parity, sum-mod- m for various m) to map identifiability as a function of operator algebra; (c) compare our compact marginalization against DeepProbLog and Scallop on the *same* grounding metric; and (d) study an active-learning loop that requests the minimum labeled symbols needed to break a detected shortcut.

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